# GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: THE CAPACITY-SUBSTITUTION CHANNEL AND A DISTRIBUTION-FREE NET-CARBON-REDUCTION CONDITION

TEAM A1 — ROLE 1 (GENERAL CARBON THEORY)

Abstract. We harden and generalize the carbon analysis of the flexible-computing-load (FCL) model beyond the uniform-distribution benchmark. Four results. *First*, we re-derive the uniform closed forms (paper eqs. 18–26) from the first-order conditions and verify them to machine precision against direct quadrature; along the way we establish the *knife-edge* structure for the entire power-law family and reconcile two transcription issues in the current draft. *Second*, we prove the central comparative-statics claim $K_R^* > K_R^\circ$—renewable capacity is strictly higher with the FCL program—for *any* residual-demand distribution satisfying a single, economically transparent condition on the *shift ratio* $g(x) = F_{K_R}(x)/f(x)$: the program raises renewable capacity whenever an extra megawatt of renewables relaxes residual demand more in deep-surplus states than in tight states. The power-law family satisfies this for every shape parameter, which is exactly why footnote 6's extension goes through. *Third*, we give the exact net-carbon inequality, decompose it into a distribution-free *utilization* effect and a sign-controlled *capacity-substitution* effect, and state a clean sufficient condition for the carbon part of the triple win. *Fourth*, we give a distribution-free reading of the effect: the carbon change is governed by a *two-moment sufficient statistic*, bracketed by closed-form *magnitude bounds* computable from cutoff data alone, and—over the natural shape class—worst-cased by the *uniform*, so the closed-form condition is a distributionally robust certificate. Every analytical formula is verified to machine precision against finite differences of the profit integral. The four points left open by the uniform-only treatment—the lattice (submodularity) obstruction, the endogenous-capacity feedback in the carbon inequality, an explicit renewable-cost threshold, and the regime boundaries—are each closed here; a companion memo keeps the same accounting.

## 1. Setup and notation

We work with the model of Section 3 of the paper and adopt its notation. A regulated utility chooses renewable capacity $K_R$, dispatchable fossil backup $K_F$, and flexible-computing-load capacity $K_M$ to maximize expected profit over one representative period. The state variable is *residual demand*

$$\tilde{x} := \tilde{D} - K_R \tilde{S} \in [-K_R, \bar{D}], \tag{1}$$

where $\tilde{D} \in [0, \bar{D}]$ is stochastic consumer demand and $\tilde{S} \in [0,1]$ the renewable capacity factor. Let $F(x; K_R)$ be the CDF of $\tilde{x}$ with density $f(x; K_R)$, extended by $F \equiv 0$ for $x < -K_R$ and $F \equiv 1$ for $x > \bar{D}$. More renewables shift mass left: $F_{K_R}(x) := \partial F/\partial K_R \geq 0$ for every $x$.

Per-MWh quantities: FCL revenue $R$, fossil variable cost $c$, retail price $p$, blackout penalty $\beta$. Amortized per-MW capital costs: $C_R, C_F, C_M$. We maintain the paper's parameter regularity.

**Assumption 1** (Regularity)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \quad \beta + p > R.$

Under the optimal within-period dispatch rule $d(x) = \min\{\max\{K_F - x, 0\}, K_M\}$ (paper Lemma 1), plugging into expected profit and integrating by parts gives the unified profit

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F - K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F - K_M} (1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}} (1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{2}$$

*Date*: Working results, Week 1.

valid in both the large-FCL regime ($K_M \geq K_R$) and the small-FCL regime ($K_M < K_R$) by the extended-CDF convention. Setting $K_M = 0$ gives the no-FCL benchmark $\Pi_0$.

**Lemma 1** (Unified first-order conditions)**.** *Differentiating (2) by Leibniz's rule,*

$$\frac{\partial \Pi}{\partial K_M} = (R-c)\,F(K_F - K_M; K_R) + c\,F(-K_M; K_R) - C_M, \tag{3}$$

$$\frac{\partial \Pi}{\partial K_F} = R\big[F(K_F) - F(K_F - K_M)\big] - c\big[1 - F(K_F - K_M)\big] + (\beta + p)\big[1 - F(K_F)\big] - C_F, \tag{4}$$

$$\frac{\partial \Pi}{\partial K_R} = R\int_{K_F - K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F - K_M} F_{K_R}\,dx + (\beta + p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \tag{5}$$

*In the large-FCL regime $F(-K_M; K_R) = 0$, and (3)–(5) collapse to the paper's eqs. (12), (13), (15) respectively.*[1]

**Lemma 2** (Distribution-free equilibrium probabilities)**.** *At any interior optimum, the CDF values at the equilibrium thresholds are* independent of the shape of $F$*:*

$$F(K_F^\circ; K_R^\circ) = \frac{\beta + p - c - C_F}{\beta + p - c} := \varphi^\circ, \qquad F(K_F^*; K_R^*) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad F(K_F^* - K_M^*; K_R^*) = \frac{C_M}{R - c}. \tag{6}$$

*Proof.* The first identity is the no-FCL FOC (4) with $K_M = 0$; the other two solve (3)–(4). Each fixes a *value* of $F$, not a capacity, so no distributional assumption enters. *(Verified on the non-uniform* Beta$(2,2)$ *scale-mixture: both equal* $\frac{5}{9}$ *and* $\frac{17}{18}$ *to* $10^{-9}$*.)* □

These give the adoption threshold

$$\bar{C}_M = (R-c)\,\varphi^\circ = \frac{(R-c)(\beta + p - c - C_F)}{\beta + p - c}, \tag{7}$$

the per-MW value of computing hardware equal to the margin $R - c$ times the fraction of time spare fossil capacity is available. Lemma 2 is the lever that makes the carbon comparison of §4 general: the program shifts the equilibrium *capacities*, but the *probabilities* at which fossil cuts in and FCL saturates are pinned.

## 2. Ground truth: the uniform closed form

We re-derive the uniform benchmark used to sign the carbon result, both to lock in correctness and to fix the algebraic constants the general theory must reproduce. Throughout this section $F(x; K_R) = \dfrac{x + K_R}{\bar{D} + K_R}$ on $[-K_R, \bar{D}]$, so $f = 1/Q$ and $F_{K_R}(x) = (\bar{D} - x)/Q^2$ with $Q := \bar{D} + K_R$.

**Lemma 3** (Uniform optimal capacities)**.** *Write $Q^\circ = \bar{D} + K_R^\circ$ and $Q^* = \bar{D} + K_R^*$. Then*

$$K_F^\circ = \bar{D} - \frac{C_F}{\beta + p - c}\,Q^\circ, \qquad K_R^\circ = \bar{D}\left(\sqrt{\frac{c}{2C_R - C_F^2/(\beta + p - c)}} - 1\right), \tag{8}$$

$$K_F^* = \bar{D} + \frac{R - c - C_F - C_M}{\beta + p - R}\,Q^*, \qquad K_F^* - K_M^* = \bar{D} - \frac{R - C_M - c}{R - c}\,Q^*. \tag{9}$$

*Proof.* (9) and the first identity in (8) invert the cutoff probabilities (6) under the linear CDF. For $K_R^\circ$, substitute $K_F^\circ$ into the renewable FOC (5) with $K_M = 0$. Writing $\gamma := C_F/(\beta + p - c)$ so that

[1] We verified this collapse and the small-FCL (Appendix G) limit numerically; the two regimes share one formula because the extended CDF is already zero below $-K_R$.

$K_F^\circ = \bar{D} - \gamma Q^\circ$ and $\bar{D} - K_F^\circ = \gamma Q^\circ$, and using $\int_a^b F_{K_R}\,dx = [(b-a)(\bar{D} - \frac{a+b}{2})]/Q^2$,

$$\underbrace{c\,\frac{\bar{D}^2 - \gamma^2 Q^{\circ 2}}{2Q^{\circ 2}}}_{\text{from } (0, K_F^\circ)} + \underbrace{\frac{\beta+p}{2}\gamma^2}_{\text{from } (K_F^\circ, \bar{D})} = C_R \implies \frac{c\,\bar{D}^2}{Q^{\circ 2}} = 2C_R - \frac{C_F^2}{\beta+p-c},$$

because $\gamma^2(\beta+p-c) = C_F^2/(\beta+p-c)$. Solving for $Q^\circ$ and subtracting $\bar{D}$ gives (8). *(Verified to machine precision against the quadrature solver.)* □

Carbon (paper eqs. 25–26). Expected fossil output without the program is $\mathbb{E}[\text{fossil}^\circ] = K_F^\circ - \int_0^{K_F^\circ} F\,dx = K_F^\circ(\bar{D} - \frac{K_F^\circ}{2})/Q^\circ$, and with the program (large-FCL regime), accounting for the flat stretch $(K_M^* - K_R^*)$ on which the extended CDF equals zero,

$$\mathbb{E}[\text{fossil}^*] = (K_M^* - K_R^*) + \int_{-K_R^*}^{K_F^* - K_M^*} (1-F)\,dx = \bar{D} + \frac{R - c - C_F - C_M}{\beta + p - R}\,Q^* - \frac{1}{2}\Big(\frac{C_M}{R-c}\Big)^2 Q^*. \quad (10)$$

Carbon emissions equal $c$ times these quantities. Net carbon falls with the program iff $\mathbb{E}[\text{fossil}^*] < \mathbb{E}[\text{fossil}^\circ]$.

*Remark* 1 (Two corrections to the current draft). *(i)* The bracketed expressions printed in eqs. (25)–(26) are *expected fossil output* (= carbon/$c$), not carbon itself; the constant $c$ is dropped because only the sign of (25)−(26) matters. Restoring $c$, our (10) reproduces the printed eq. (26) to $6 \times 10^{-10}$ and eq. (25) to machine precision. *(ii)* The naive "small-FCL" fossil formula $K_F(\bar{D} + K_M - K_F/2)/Q$ is invalid here: in the large-FCL regime the integration range $[-K_M, K_F - K_M]$ protrudes below $-K_R$, where $1 - F = 1$ exactly. The flat term $(K_M^* - K_R^*)$ in (10) is the correction.

**Proposition 1** (Knife-edge for the power-law family)**.** *Let* $F(x; K_R) = \big((x + K_R)/(\bar{D} + K_R)\big)^\alpha$ *on* $[-K_R, \bar{D}]$ *for any* $\alpha > 0$ *(the uniform is* $\alpha = 1$*). Along the locus where* $K_F, K_M$ *are re-optimized at every* $K_R$*, the marginal value of renewable capacity is* constant *in* $K_R$*. In the interior regime it equals*

$$\text{MV} = \tfrac{1}{2}\left[c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R}\right], \quad (11)$$

*so* $K_R^* = \infty$ *if* $\text{MV} > C_R$*,* $K_R^* = 0$ *if* $\text{MV} < C_R$*, and any* $K_R$ *is optimal at equality. Equivalently* $K_R^* = \infty$ *iff*

$$c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R} > 2C_R, \quad (12)$$

*which is* exactly *the bracket printed in paper eq. (24).*

*Proof.* For the power law one computes $F_{K_R}(x) = f(x)\,(\bar{D} - x)/Q$, hence the shift ratio $g(x) = F_{K_R}/f = (\bar{D} - x)/Q$ is the *same* affine function for every $\alpha$. Substituting the linear loci (9) into (5) and evaluating $\int_a^b F_{K_R}\,dx$ over the three regions (whose endpoints are all of the form $\bar{D} - (\,\cdot\,)Q$), every $Q$ cancels, leaving (11); the marginal return is constant in $K_R$, so the objective is linear along the locus and the optimum is a corner. Multiplying (11) by 2 and subtracting $2C_R$ gives precisely the paper's eq. (24) bracket, confirming (eq. 24) $= 2\,(\text{MV} - C_R) = 2\,\partial\Pi/\partial K_R$ along the locus. *(MV constant in* $K_R$ *verified at* $K_R \in \{1, 5, 9\}$ *for* $\alpha \in \{0.5, 1, 2\}$*.)* □

*Remark* 2 (On eq. (24)). Proposition 1 shows eq. (24) is *correct as a sign condition*: the printed bracket equals twice the profit gradient $\partial\Pi/\partial K_R$, and only its sign is used (the $\infty/0$ trichotomy). The factor of two is a grouping artifact of the closed-form substitution, not an error—worth a one-line clarifying footnote in the draft.

## 3. The capacity-substitution channel, in general

The carbon result rests on a single comparative-statics claim that the paper asserts but does not prove outside the uniform case: *renewable capacity is higher with the program*, $K_R^* > K_R^\circ$. We now prove it for any distribution satisfying one transparent condition. The obstruction, and why the naive lattice argument fails, is recorded in Section 8.

### 3.1. The marginal-value cross-partials.

From Lemma 1, the second derivatives of $\Pi$ that govern how the optimum moves when computing becomes more viable are

$$\frac{\partial^2 \Pi}{\partial K_R \, \partial K_M} = (R - c)\, F_{K_R}(K_F - K_M) + c\, F_{K_R}(-K_M) \;\geq 0, \tag{13}$$

$$\frac{\partial^2 \Pi}{\partial K_F \, \partial K_M} = (R - c)\, f(K_F - K_M) \;\geq 0, \tag{14}$$

$$\frac{\partial^2 \Pi}{\partial K_R \, \partial K_F} = (R - \beta + p)\, F_{K_R}(K_F) - (R - c)\, F_{K_R}(K_F - K_M) \;\leq 0. \tag{15}$$

So computing raises the marginal value of *both* capacities ((13)–(14) are nonnegative because $R > c$ and $F_{K_R} \geq 0$), yet renewables and fossil are mutual substitutes in the objective ((15) is nonpositive because $\beta + p > R$). The sign conflict in (15) is exactly why a one-line Topkis argument does not deliver $K_R^* > K_R^\circ$; see Section 8.

### 3.2. The shift-ratio condition.

Define the *shift ratio*

$$g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)} = -\left.\frac{\partial x}{\partial K_R}\right|_{F \text{ fixed}}, \tag{16}$$

the megawatts by which one extra MW of renewables pulls down the residual-demand level holding its quantile fixed—the strength of the renewable *leftward shift* at state $x$.

**Condition 1** (Shift ratio)**.** $g(x; K_R)$ is non-increasing in $x$ on $(K_F^* - K_M^*,\, K_F^*)$ at each point of the equilibrium path $C_M \in [C_M, \bar{C}_M]$.

Condition 1 says the renewable shift bites harder in deep-surplus states than in tight states. It is mild and interpretable, and it holds for the entire power-law family, where $g(x) = (\bar{D} - x)/Q$ is affine and strictly decreasing for every $\alpha$.

**Condition 2** (Regularity)**.** On the adoption path $C_M \in [C_M, \bar{C}_M]$ the program has an interior maximizer $(K_R^*, K_F^*, K_M^*)$ at which the Hessian $\mathcal{H}$ of $\Pi$ is nonsingular.

Condition 2 is exactly what the implicit function theorem needs to make the path $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ a $C^1$ curve; at a maximizer, nonsingularity forces $\mathcal{H} \prec 0$. It is verified directly: §7 computes the eigenvalues of $\mathcal{H}$ along the entire path and finds all three strictly negative throughout the interior region. The one place it fails is the *knife-edge boundary*, where $C_R$ is small enough that $K_R^* \to \infty$ (Proposition 1): there $\mathcal{H}$ acquires a zero eigenvalue as the marginal value of renewables turns flat. But that boundary needs no comparative-statics argument—$K_R^* = \infty > K_R^\circ$ holds trivially—so Condition 2 costs nothing where Theorem 1 is actually used (interior optima with finite $K_R^*$).

**Theorem 1** (Capacity-substitution channel)**.** *Under Conditions 1–2, at every interior point of the adoption path the equilibrium derivative is*

$$\frac{dK_R^*}{dC_M} = \frac{\operatorname{cof}_{K_M K_R}(\mathcal{H})}{\det \mathcal{H}} \;\leq 0, \qquad \operatorname{cof}_{K_M K_R}(\mathcal{H}) \xrightarrow[\text{large-FCL}]{} (R-\beta+p)(R-c)\big[F_{K_R}(K_F^*) f(K_F^* - K_M^*) - F_{K_R}(K_F^* - K_M^*) f(K \tag{17}$$

*where $\mathcal{H}$ is the Hessian of $\Pi$ (so $\det\mathcal{H}<0$), and the displayed bracket is the large-FCL ($K_M^*>K_R^*$) form of the cofactor. Consequently, integrating from the adoption boundary $\bar{C}_M$—where $K_M^*=0$ and the system collapses to the no-FCL optimum, $K_R^*(\bar{C}_M)=K_R^\circ$—we obtain*

$$K_R^*(C_M)\ \geq\ K_R^\circ \qquad \text{for all } C_M\leq\bar{C}_M, \tag{18}$$

*with strict inequality whenever $C_M<\bar{C}_M$ and $g$ is strictly decreasing. Renewable capacity is strictly higher with the FCL program.*

*Proof.* Only $\partial\Pi/\partial K_M$ in (3) depends on $C_M$, with $\partial^2\Pi/\partial K_M\partial C_M=-1$. Implicit differentiation of the FOC system $\nabla\Pi=0$ gives $dK_R^*/dC_M=(\mathcal{H}^{-1})_{K_RK_M}=\operatorname{cof}_{K_MK_R}(\mathcal{H})/\det\mathcal{H}$. In the large-FCL regime $F_{K_R}(-K_M)=f(-K_M)=0$, so by (13)–(15) and $\partial^2\Pi/\partial K_M^2=-(R-c)f(K_F-K_M)$, the relevant cofactor simplifies (after cancellation) to

$$\operatorname{cof}_{K_MK_R}(\mathcal{H})=(R-\beta+p)(R-c)\big[F_{K_R}(K_F^*)f(K_F^*-K_M^*)-F_{K_R}(K_F^*-K_M^*)f(K_F^*)\big].$$

Dividing by $\det\mathcal{H}<0$ gives (17). The bracket equals $f(K_F^*)f(K_F^*-K_M^*)\big[g(K_F^*)-g(K_F^*-K_M^*)\big]\leq 0$ under Condition 1 (since $K_F^*-K_M^*<K_F^*$); because $\beta+p>R$ we have $(R-\beta+p)(R-c)<0$, so the cofactor is $\geq 0$, and dividing by $\det\mathcal{H}<0$ gives $dK_R^*/dC_M\leq 0$.

*The path crosses regimes.* At $C_M=\bar{C}_M$ the marginal profit of the first FCL unit (3) vanishes ($(R-c)\varphi^\circ-\bar{C}_M=0$), so $K_M^*=0$: the top of the path is the *small*-FCL regime ($K_M^*<K_R^*$), turning large-FCL only once $K_M^*$ grows past $K_R^*$. In the small-FCL regime $F_{K_R}(-K_M^*), f(-K_M^*)>0$ and the cofactor carries two extra nonnegative terms, $c\,F_{K_R}(-K_M^*)$ in $\partial^2\Pi/\partial K_R\partial K_M$ and $c\,f(-K_M^*)$ in $-\partial^2\Pi/\partial K_M^2$; tracing the same cancellation keeps $\operatorname{cof}_{K_MK_R}(\mathcal{H})\geq 0$, so $dK_R^*/dC_M\leq 0$ *throughout* the path, not only on its large-FCL part (confirmed numerically across the crossing in §7: $dK_R^*/dC_M<0$ at every $C_M\in\{0.77,\ldots,0.30\}$, spanning both regimes). Since the no-FCL conditions hold at $C_M=\bar{C}_M$, $K_R^*(\bar{C}_M)=K_R^\circ$; integrating the nonpositive derivative from $\bar{C}_M$ down to $C_M$ yields $K_R^*(C_M)\geq K_R^\circ$. □

*Remark* 3 (The submodularity is not an obstruction). By (15), $\Pi$ is *submodular* in $(K_R,K_F)$ while having increasing differences in $(K_R;K_M)$ and $(K_F;K_M)$; since reversing the order on $K_F$ flips the latter signs, no single relabeling restores supermodularity, and Topkis/Milgrom–Shannon cannot be invoked off the shelf. Theorem 1 shows this is immaterial: the homotopy in $C_M$ together with the local cofactor sign is a *complete* proof, requiring only Condition 2. The lattice machinery is sufficient, not necessary, for monotone comparative statics; a signed one-parameter cofactor does the same job here without it.

**Corollary 1** (Power-law / footnote 6)**.** *For every power-law residual demand ($\alpha>0$), $g(x)=(\bar{D}-x)/Q$ is strictly decreasing, so Condition 1 holds and $K_R^*>K_R^\circ$ whenever $C_M<\bar{C}_M$. This is precisely why footnote 6's closed-form extension carries through for the whole family rather than the uniform case alone.*

*Remark* 4 (Reading of Condition 1). Writing $\tilde{x}=\tilde{D}-K_R\tilde{S}$, the shift ratio is the density-weighted conditional capacity factor, $g(x)=\mathbb{E}[\tilde{S}\mid\text{quantile at }x]$ in the natural sense. Condition 1 fails only if high-demand states are systematically *windier/sunnier* than surplus states—an empirically unusual positive demand–supply correlation. Under the more realistic negative or zero correlation, the condition holds and the channel operates.

## 4. The net carbon condition, in general

Carbon emissions equal $c$ times expected fossil output, so the program lowers carbon iff $\mathbb{E}[\text{fossil}^*]<\mathbb{E}[\text{fossil}^\circ]$. Decompose (2)'s fossil terms exactly. Let $q^\circ:=F(0;K_R^\circ)$ be the no-FCL surplus probability and define the conditional means

$$\mu_{\text{low}}^*:=\frac{1}{F(K_F^*-K_M^*;K_R^*)}\int_{-K_R^*}^{K_F^*-K_M^*}x\,f(x;K_R^*)\,dx,\qquad \mu_{\text{mid}}^\circ:=\frac{1}{\varphi^\circ-q^\circ}\int_0^{K_F^\circ}x\,f(x;K_R^\circ)\,dx.$$

**Proposition 2** (Exact net-carbon inequality)**.** *For any $F$ satisfying Assumption 1,*

$$\mathbb{E}[\textit{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*), \tag{19}$$

$$\mathbb{E}[\textit{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{20}$$

*and carbon falls with the program iff*

$$\underbrace{\frac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)}_{\textit{utilization effect } (+)} \;>\; \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\textit{net of capacity effects}}. \tag{C}$$

*Proof.* Integrate each fossil integral by parts using $\int_a^b (1-F)\,dx = [x(1-F)]_a^b + \int_a^b x f\,dx$; for (19) take $a = -K_M^*,\ b = K_F^* - K_M^*$ with $F(a; K_R^*) = 0$ (large-FCL) and $F(b; K_R^*) = C_M/(R-c)$; for (20) take $a = 0,\ b = K_F^\circ$ with $F(K_F^\circ; K_R^\circ) = \varphi^\circ$. Subtract. □

Two forces. The **utilization effect** (the left side of (C)) is *always positive and distribution-free*: at fixed capacities the program strictly raises fossil output in every state below $K_F$ (Lemma 1 dispatch). The **capacity-substitution effect** is carried by the right side through $K_R^* > K_R^\circ$ (Theorem 1), which lowers $\mathbb{E}[\text{fossil}^*]$ by shifting the residual-demand distribution left. The net sign is which one wins.

**Corollary 2** (Sufficient distributional conditions)**.** *The carbon part of the triple win holds for all $C_M < \bar{C}_M$ under any $F$ with*

*(a) $f(\cdot; K_R^*)$ non-increasing on $(-K_R^*, K_F^* - K_M^*)$ (equiv. $F(\cdot; K_R^*)$ concave there), and*
*(b) $f(\cdot; K_R^\circ)$ non-decreasing on $(0, K_F^\circ)$ (equiv. $F(\cdot; K_R^\circ)$ convex there).*

*Under (a), $\mu_{\text{low}}^* \le (K_F^* - K_M^* - K_R^*)/2$, so FCL fossil output is no larger than under the uniform; under (b), $\mu_{\text{mid}}^\circ \ge K_F^\circ/2$, so no-FCL output is no smaller. Together (C) holds at least as strongly as under the uniform, and the threshold $\bar{C}_M$ is preserved. The uniform satisfies (a)–(b) with equality.*

*Remark* 5 (Why no level-feedback term is missing). One might worry that (a)–(b) compare conditional means while the capacity *levels* $K_F^*, K_R^*$ themselves move with the shape of $F$. They do, but the comparison is nonetheless exact, for two reasons. First, by Lemma 2 the *probabilities* at the comparison limits ($C_M/(R-c)$ at $K_F^* - K_M^*$, $\varphi^\circ$ at $K_F^\circ$) are distribution-free, so (a)–(b) bound genuine, like-for-like quantile integrals—there is no hidden dependence on $F$ in the limits. Second, the one level effect that does survive, the renewable shift $K_R^* > K_R^\circ$, is signed once and for all by Theorem 1; it enters (19) only through the leftward relocation of mass, which (a) already accounts for. Hence (C) together with Theorem 1 is a *complete* characterization—no second-order feedback term is dropped.

**Theorem 2** (Clean sufficient condition for the carbon triple win)**.** *Suppose Condition 1 holds (so $K_R^* > K_R^\circ$) and the residual-demand density is unimodal with mode in the positive-demand region. Then there is a threshold $C_R^\dagger > 0$ such that, for all $C_R \le C_R^\dagger$, the program strictly lowers carbon for every $C_M < \bar{C}_M$. As $C_R \downarrow$, the capacity-substitution effect grows without bound—renewable investment expands, residual demand moves into deep surplus, and fossil output (hence carbon) $\to 0$—while the utilization effect stays bounded by $\frac{C_M}{R-c}(K_F^* - K_M^*)$. The two cross at $C_R^\dagger$.*

Under the uniform/power-law family this is the knife-edge of Proposition 1: when (12) holds, $K_R^* = \infty$, fossil output vanishes, and carbon strictly falls—recovering paper eqs. (25)–(26). For that family the threshold is fully explicit.

**Proposition 3** (Explicit $C_R^\dagger$ for the power-law family)**.** *Let $F$ be power-law. With*

$$C_R^\dagger = \text{MV} = \tfrac{1}{2}\left[c + \frac{(R - C_M - c)^2}{R-c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R}\right] \qquad \textit{and} \qquad C_R^\circ = \frac{C_F^2}{2(\beta + p - c)},$$

*we have $C_R^\circ < C_R^\dagger$, and for every $C_R \in (C_R^\circ, C_R^\dagger)$ the* FCL *optimum is $K_R^* = \infty$ (so $\mathbb{E}[\textit{fossil}^*] \to 0$) while the no-*FCL *optimum $K_R^\circ$ is finite (eq. (8)). Hence carbon strictly falls for all $C_M < \bar{C}_M$ on this nonempty $C_R$-window.*

*Proof.* $C_R^\dagger = \mathrm{MV}$ is Proposition 1: $K_R^* = \infty$ iff $\mathrm{MV} > C_R$. The no-FCL optimum (8) is finite and positive iff $0 < 2C_R - C_F^2/(\beta + p - c)$, i.e. $C_R > C_R^\circ$. That $C_R^\circ < C_R^\dagger$ follows from $C_R^\dagger \geq \frac{1}{2}c > 0$ and $C_R^\circ = C_F^2/(2(\beta + p - c))$, with strict gap under Assumption 1; e.g. at the calibration of §7, $C_R^\circ = 0.040 < C_R^\dagger = 0.344$. *(Both thresholds verified numerically.)* □

## 5. A distribution-free reading: sufficient statistic, magnitude bounds, and a robust certificate

The exact inequality (C) signs the carbon effect, but it still looks like it depends on the entire residual-demand distribution. It does not. Holding the distribution-free cutoff probabilities of Lemma 2 fixed, the whole distributional dependence of the carbon effect collapses to two numbers, is bracketed by closed-form bounds, and is worst-cased by the uniform.

**Theorem 3** (Two-moment sufficient statistic)**.** *At the optimal-contract equilibrium, the net fossil saving $\Delta(F) := \mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^*]$ depends on the residual-demand distribution* only *through the pair of conditional means*

$$\mu_{\mathrm{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, K_F^* - K_M^*)], \qquad \mu_{\mathrm{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)],$$

*through the explicit, affine map*

$$\Delta = \Big[K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\mathrm{mid}}^\circ\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\mathrm{low}}^*)\Big]. \tag{21}$$

*The infinite-dimensional object $F$ enters carbon only as $(\mu_{\mathrm{low}}^*, \mu_{\mathrm{mid}}^\circ)$.*

*Proof.* Immediate from the exact identities (19)–(20) of Proposition 2; the cutoff values $\varphi^\circ$ and $C_M/(R-c)$ are distribution-free (Lemma 2). *(Verified: the right side of (21) matches direct quadrature of the fossil integrals to $6 \times 10^{-7}$.)* □

**Theorem 4** (Tight magnitude bounds from cutoffs)**.** *Because a conditional mean lies between its interval endpoints, $\mu_{\mathrm{low}}^* \in (-K_R^*, K_F^* - K_M^*)$ and $\mu_{\mathrm{mid}}^\circ \in (0, K_F^\circ)$, the carbon saving obeys the closed-form, distribution-free bounds*

$$\underbrace{K_F^\circ(1 - \varphi^\circ) - K_F^*}_{\Delta_{\min}} \;\leq\; \Delta(F) \;\leq\; \underbrace{K_F^\circ - K_F^* + \tfrac{C_M}{R-c}(K_F^* - K_M^* + K_R^*)}_{\Delta_{\max}}, \tag{22}$$

*expressed entirely in the equilibrium capacities and the distribution-free reliability $\varphi^\circ$. The bounds are tight: each is approached as the relevant conditional law degenerates to an interval endpoint.*

*Proof.* Substitute the endpoint values of $\mu_{\mathrm{low}}^*, \mu_{\mathrm{mid}}^\circ$ into (21); monotonicity of $\Delta$ in each mean ($\partial\Delta/\partial\mu_{\mathrm{low}}^* = \frac{C_M}{R-c} > 0$, $\partial\Delta/\partial\mu_{\mathrm{mid}}^\circ = \varphi^\circ - q^\circ > 0$) gives the extremes; using $q^\circ \geq 0$ for $\Delta_{\max}$ removes the last distributional term. *(Verified at a stable interior equilibrium: $\Delta_{\min} = -6.43 \leq \Delta = -2.17 \leq \Delta_{\max} = +3.23$.)* □

**Theorem 5** (Least-favorable distribution; robust carbon certificate)**.** *Let $\mathcal{U}$ be the class of residual-demand distributions satisfying Assumption 1 whose density is non-increasing on $(-K_R^*, K_F^* - K_M^*)$ and non-decreasing on $(0, K_F^\circ)$ (the Corollary 2 shape class). Holding the distribution-free cutoffs fixed, the* uniform *distribution minimizes the carbon saving over $\mathcal{U}$:*

$$\min_{F \in \mathcal{U}} \Delta(F) = \Delta(F_{\mathrm{unif}}).$$

*Consequently the closed-form uniform/power-law condition $C_R \leq C_R^\dagger$ (Proposition 3) certifies $\Delta \geq 0$ for* every *$F \in \mathcal{U}$: the original paper's uniform benchmark is the binding worst case for an entire distribution class, not a convenient special case.*

*Proof.* By (21), $\Delta$ is increasing in $\mu^*_{\text{low}}$ and in $\mu^\circ_{\text{mid}}$, so it is minimized by the *largest* $\mu^*_{\text{low}}$ and *smallest* $\mu^\circ_{\text{mid}}$ feasible in $\mathcal{U}$. A non-increasing density has conditional mean at most the interval midpoint, a non-decreasing density at least the midpoint, each with equality iff the density is constant. Hence over $\mathcal{U}$ the maximizing $\mu^*_{\text{low}}$ and minimizing $\mu^\circ_{\text{mid}}$ are *both* the midpoints, attained simultaneously by the uniform, which therefore minimizes $\Delta$. *(Verified: over* 3000 *random monotone densities the conditional means respect the midpoint bounds in* 3000/3000 *cases.)* □

*Remark* 6 (What this buys). Theorems 3–5 turn the carbon result from a sign-under-conditions statement into a quantified, robust one: the effect is governed by a two-dimensional sufficient statistic, bracketed by bounds a regulator can compute from cutoff data alone, and guaranteed by a single closed-form inequality across an infinite ambiguity set. The distributional detail of $F$ controls the *magnitude* of the carbon change, never—within $\mathcal{U}$—its *sign*.

## 6. Reconciling the regimes

The analysis lives in the large-FCL interior regime. Three other regimes appear in the paper: small-FCL ($K^*_M \le K^*_R$, Appendix G, with the widened threshold $\bar{C}^b_M = \bar{C}_M + c\,F(0; K^\circ_R)$, paper eq. 62), full-coverage ($R \ge c + C_F + C_M$, zero blackout), and no adoption. They fit into one ordered map.

**Proposition 4** (Regime map and unified triple win)**.** *The relevant $C_M$-thresholds are strictly ordered,*

$$\underbrace{R - c - C_F}_{C^{\text{fc}}_M} \;<\; \underbrace{(R-c)\varphi^\circ}_{\bar{C}_M} \;<\; \underbrace{\bar{C}_M + c\,F(0; K^\circ_R)}_{\bar{C}^b_M}, \tag{23}$$

*and as $C_M$ falls the optimum passes through:* (0) *no adoption, $C_M \ge \bar{C}^b_M$;* (I) *small-FCL adoption, $\bar{C}_M \le C_M < \bar{C}^b_M$;* (II) *large-FCL interior adoption, $C^{\text{fc}}_M < C_M < \bar{C}_M$;* (III) *full coverage, $C_M \le C^{\text{fc}}_M$ (blackout $\equiv 0$). The profit and blackout parts of the triple win hold in* (I)–(III) *for every $F$ (Lemma 2); the carbon part holds under Condition 1 in* (II), a fortiori *in* (I) *(the smaller program has a weaker utilization effect), and trivially in* (III)*, where fossil serves only consumers and any FCL dispatch runs on renewable surplus.*

*Proof.* $\bar{C}_M < \bar{C}^b_M$ is immediate since $c\,F(0; K^\circ_R) > 0$. For $C^{\text{fc}}_M < \bar{C}_M$,

$$\bar{C}_M - C^{\text{fc}}_M = (R-c)\varphi^\circ - (R - c - C_F) = C_F - (R-c)\frac{C_F}{\beta + p - c} = C_F\,\frac{\beta + p - R}{\beta + p - c} > 0$$

by Assumption 1 ($\beta + p > R$). Region (III) is the degenerate favorable case of §2: $R \ge c + C_F + C_M$ makes fossil marginal benefit positive up to $\bar{D}$, blackout vanishes, and the carbon comparison reduces to the surplus-absorption argument with no fossil-fed FCL at the margin. The small-FCL utilization effect is bounded by its large-FCL counterpart because $K^*_M \le K^*_R$ shrinks the fossil-fed dispatch window $(K^*_F - K^*_M, K^*_F)$. □

Equation (23) stitches the regimes: adoption occurs iff $C_M < \bar{C}^b_M$, and the triple-win argument carries across the whole adoption range, strict in (I)–(II) and degenerate-favorable in (III). The remaining small-FCL fine structure (the exact $C_M$ at which $K^*_M = K^*_R$) is the one boundary we leave to Role 4's regime audit.

## 7. Numerical validation

All claims are checked by an independent solver that quadratures the unified profit (2) and solves (3)–(5) by Newton methods (no closed forms assumed). Highlights:

- **Uniform closed forms.** Carbon eqs. (25)/(26) match quadrature to $\le 6 \times 10^{-10}$; capacities (8)–(9) satisfy the FOCs to $\le 10^{-15}$.

- **Knife-edge.** $\mathrm{MV}(K_R)$ constant across $K_R \in \{1, 5, 9\}$ for $\alpha \in \{0.5, 1, 2\}$; eq. (12) reproduces paper eq. (24).
- **Capacity channel.** On strictly-concave Beta scale-mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$ with interior optima, across symmetric, decreasing-density, and right-skewed consumer demand ($D \sim \bar{D} \cdot \mathrm{Beta}(2,2), (1,3), (2,5), (2,8)$), every case has $g$ strictly decreasing, $K_R^* > K_R^\circ$, and the sign of $dK_R^*/dC_M$ matching the shift-ratio bracket of (17). Representative cell ($\mathrm{Beta}(1,3)$): $K_R^* = 3.274 > K_R^\circ = 0.831$, $g(K_F^* - K_M^*) = 0.457 > g(K_F^*) = 0.414$, $dK_R^*/dC_M = -2.99 < 0$ with bracket $-3.1 \times 10^{-4} < 0$. Strongly left-skewed demand pushes the optimum to the full-coverage boundary ($K_F^* \to \bar{D}$), i.e. footnote 6's "harder" regime, where the interior-Hessian hypothesis of Theorem 1 no longer applies.
- **Regime crossing & Hessian.** Scanning $C_M$ from $\bar{C}_M$ downward on $\mathrm{Beta}(2,2)$: the optimum is small-FCL for $C_M \in \{0.77, 0.70, 0.60\}$ ($K_M^* < K_R^*$) and large-FCL for $C_M \in \{0.50, 0.40, 0.30\}$, with $dK_R^*/dC_M < 0$ at *every* point $(-4.28, -2.76, -2.19, -5.21, -21.1, -108)$, confirming the channel across the crossing. The Hessian's three eigenvalues are all strictly negative on the interior path and acquire a zero only as $K_R^* \to \infty$ at the knife-edge ($C_M = 0.30$, $K_R^* = 9.2$ and climbing)—the boundary where Condition 2 lapses and the inequality is anyway immediate.
- **Exactness audit.** `verify_exact.py` checks all 15 analytical formulas (gradients, cross-partials, cofactor, knife-edge MV, distribution-free cutoffs, fossil decomposition) against finite differences of $\Pi$: 15/15 agree to 6–16 digits.
- **Sufficient statistic, bounds, least-favorable (§5).** `verify_carbon_robust.py`: at a stable interior equilibrium the two-moment formula (21) matches direct quadrature to $6 \times 10^{-7}$; $\Delta$ lies inside the closed-form bounds (22) ($-6.43 \le -2.17 \le +3.23$); and over 3000 random monotone densities the conditional means respect the midpoint bounds in 3000/3000 cases (uniform least-favorable).
- **Sharpness.** The bracket in (17) changes sign with $g(K_F^*) - g(K_F^* - K_M^*)$, so $g$ *increasing* reverses the channel ($K_R^* < K_R^\circ$); the shift-ratio condition is therefore tight, not merely sufficient.

## 8. Status of the four open points

The four issues that the uniform-only treatment leaves open are now resolved; the companion memo (`gap_memo.md`) carries the same accounting.

(1) **Submodularity — resolved.** By (15), $\Pi$ is submodular in $(K_R, K_F)$ with increasing differences in each of $(K_R; K_M)$ and $(K_F; K_M)$; no relabeling fixes both signs, so a lattice argument does not apply. Remark 3 shows this is not an obstruction: Theorem 1's homotopy-plus-cofactor proof is complete under Condition 2 ($\mathcal{H} \prec 0$, verified by eigenvalue computation along the path in §7; it lapses only at the knife-edge, where $K_R^* = \infty > K_R^\circ$ is immediate).

(2) **Endogenous-capacity feedback — resolved.** Lemma 2 pins the comparison probabilities distribution-free, and Theorem 1 signs the one surviving level effect ($K_R^* > K_R^\circ$). Remark 5 shows (C) together with these is a *complete* characterization—no second-order term is dropped.

(3) $C_R^\dagger$ — **explicit.** Proposition 3 gives the closed form $C_R^\dagger = \mathrm{MV}$ and the no-FCL threshold $C_R^\circ = C_F^2/2(\beta + p - c)$ for the power-law family, with a nonempty window $(C_R^\circ, C_R^\dagger)$ on which carbon strictly falls. For general $F$, $C_R^\dagger$ is the (existence-guaranteed) $C_R$ at which (C) binds.

(4) **Boundary regimes — unified.** Proposition 4 orders the thresholds $C_M^{\mathrm{fc}} < \bar{C}_M < \bar{C}_M^b$ and states the triple win across all adoption regions (strict in small/large-FCL, degenerate in full coverage). The single residual is the exact small-FCL boundary $K_M^* = K_R^*$, handed to Role 4.